

EQUILIBRIUM

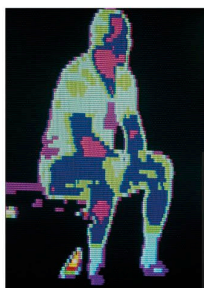
THE BODY'S CELLS AND TISSUES ONLY FUNCTION WELL IF ALL ASPECTS OF THEIR ENVIRONMENT ARE KEPT STABLE AND IN EQUILIBRIUM. SEVERAL BODY SYSTEMS MAINTAIN A BALANCED INTERNAL ENVIRONMENT, A PROCESS CALLED HOMEOSTASIS.

The biochemical reactions in cells are attuned to specific conditions, such as oxygen levels, acidity, water levels, and temperature. These must be maintained within the correct limits or the reactions go awry and the body malfunctions.

CARDIOVASCULAR SYSTEM	144-159
RESPIRATORY SYSTEM	160-175
SKIN, HAIR, AND NAILS	176-189
DIGESTIVE SYSTEM	210-239
URINARY SYSTEM	240-249

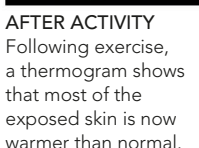
HOMEOSTATIC SYSTEMS

Several systems contribute to homeostasis. For example, the respiratory system ensures that oxygen levels are maintained; the digestive system takes in and processes nutrients; and the circulatory system distributes oxygen and nutrients and gathers waste products, which are removed by the urinary and respiratory systems.



BEFORE ACTIVITY

In this thermogram, temperature is graded from blue (cooler) to red (warmer).



AFTER ACTIVITY

Following exercise, a thermogram shows that most of the exposed skin is now warmer than normal.

CONTROL AND FEEDBACK

The body's major control systems, nerves and hormones, are mainly responsible for coordinating homeostatic mechanisms using feedback loops. For example, if water levels in the tissues fall, body fluids become more concentrated. Sensors detect this and feed back information to the brain, whose homeostatic centres

trigger regulating actions. Hormonal control of urinary excretion is adjusted to conserve water, and nervous activity produces thirst so that we drink. The sensors detect the changes as fluid concentrations return to normal, then they switch off until needed again. Thermoregulation – maintaining an approximately constant body temperature – uses the same feedback principles, with mechanisms such as sweating, blood flow to the skin, and shivering being used to regulate heat loss, conservation, and generation. In these ways, conditions inside the body are kept relatively stable, and an ongoing equilibrium is maintained.

